SOFTWARE AND HARDWARE TOOLS

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

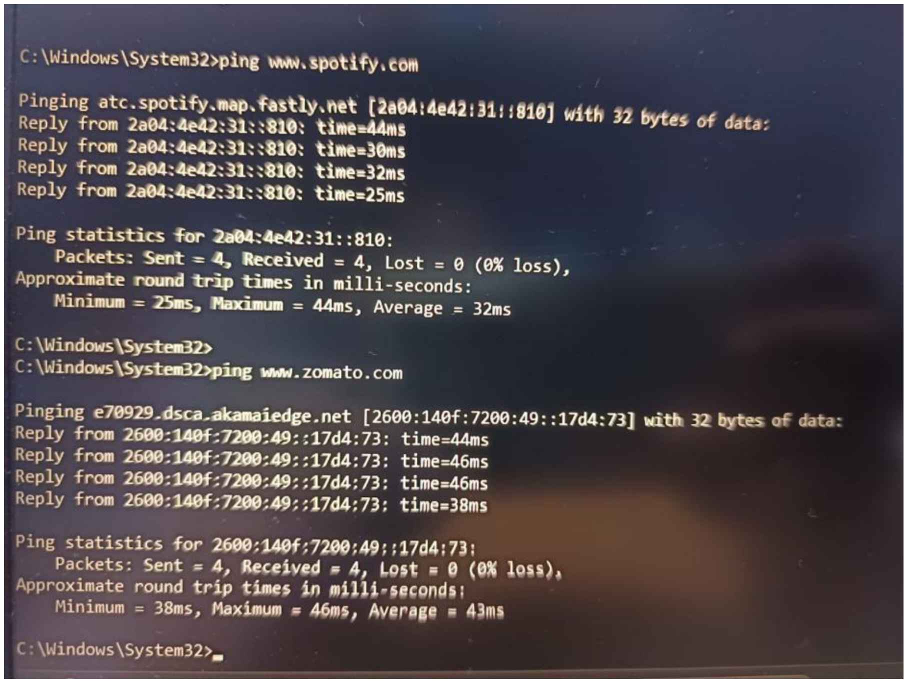
Steps

1. Install **Wireshark**.
2. Start capturing on your **Wi-Fi** or **Ethernet** interface.
3. Browse **Instagram** or **YouTube** for **2 minutes**.
4. Stop the capture.
5. Save it as **network_capture.pcap**.

Top 3 Protocols (Example)

Protocol	**Purpose**
TLS	Secure HTTPS communication
TCP	Reliable data transfer
DNS	Converts website names to IP addresses

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.



Faster Website: Spotify (example only—your results may differ).

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Windows

1. Press Ctrl + Shift + Esc
2. Open Task Manager
3. Go to Processes

Example

Process Name	Memory Usage
Chrome.exe	520 MB
Chrome.exe	120 MB

Process Name Memory Usage

WhatsApp.exe 180 MB

**4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**

Tool

- Advanced IP Scanner or Angry IP Scanner

Example Output

IP Address	Device Name
192.168.1.1	TP-Link Router
192.168.1.100	Hensie-Laptop

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Tool	**When It Is Used**
Cable Tester	Checks whether an Ethernet cable is wired correctly and detects faults.
RJ-45 Crimping Tool	Used to attach RJ-45 connectors to Ethernet cables.
Digital Multimeter	Measures voltage and checks power supply or cable continuity during troubleshooting.